

FARM MANAGEMENT SYSTEM

5. PROJECT DEVELOPMENT PHASE

5.1 Development Environment

The Farm Management System is developed as a digital platform to simplify and organize farming activities. The system provides a centralized place to maintain information about farms, crops, farming activities, resources, and production. It helps farmers reduce manual record keeping and access important farm information easily.

5.2 Custom Objects

Object	Record Name	Purpose
Farm	Farm Name (Text)	Stores basic information about the farm, including location, land area, soil type, and farming type.
Crop	Crop Name (Text)	Stores details of crops cultivated on the farm, including planting date, expected harvest date, and crop status.
Resource	Resource Name(Text)	Maintains information about farming resources such as seeds, fertilizers, pesticides, and manure.
Equipment	Equipment Name(Text)	Stores details of agricultural equipment such as tractors, pumps, harvesters, and other farming tools.
Expense	Expense Name(Text)	Records expenses related to farming activities, including labour, fertilizers, seeds, irrigation, and maintenance.
Harvest	Harvest ID(Text)	Stores information about harvested crops, including harvest date, crop name, and quantity produced.
Sales	Sales ID(Text)	Maintains details of agricultural products sold, including product, quantity, selling price, buyer, and total amount.

5.3 Farm Object Fields

Field	Type / Values
Farm Name	Text
Farm ID	Auto Number
Land Area	Number
Location	Text
Solid Type	Picklist: Sandy, Clay, Loamy, Black Soil
Farming Type	Picklist: Organic, Conventional, Mixed
Water Source	Picklist: Well, Borewell, River, Rainwater
Farm status	Picklist: Active, Inactive
Owner Name	Text
Registration Date	Date

5.4 Crop Object Fields

Field	Type / Details
Crop Name	Text
CropID	Auto Number
Crop Category	Picklist: Cereal, Vegetable, Fruit, Pulse
Seed Variety	Text
Sowing Date	Date
Expected Harvest Date	Date
Cultivated Area	Number
Crop Status	Picklist: Planned, Growing, Ready for Harvest, Harvested
Water Requirement	Picklist: Low, Medium, High
Farm	Lookup relationship to Farm

5.5 Farmer Object Fields

Field	Type / Details
Farmer Name	Text
FarmerID	Auto Number
Contact Number	Phone
Email	Email
Address	Text Area
Village	Text

District	Text
Experience	Number
Farming Experience	Picklist: Beginner, Intermediate, Experienced
Farm	Lookup relationship to Farm

5.6 Buyer Object Fields

Field	Type / Details
Buyer Name	Text
Contact Number	Phone
Email	Email
Address	Text Area
Buyer Type	Picklist: Wholesale, Retail
Product Purchased	Text
Quantity Purchased	Number
Purchase Date	Date

5.7 Lightning App

A Salesforce Lightning application named “Farm Management System” is created to manage farm-related activities in one place. The application includes Farm, Crop, Farmer, Resource, Equipment, Buyer, Reports, and Dashboards tabs. It helps users easily access and manage farming information through a single application.

5.8 Security and Access Configuration

- Admin– Has full access to the Farm Management System.
- Farm Manager– Can view and update farm and crop information.
- Farmer– Can access and manage their own farm details.
- Sales User– Can manage buyer and sales information.
- Sharing Rules– Control access to farm records between users.

5.9 Development Summary

The development phase converts the Farm Management System design into a working Salesforce application. The system manages farms, crops, farmers, resources, equipment, buyers, and sales. It helps reduce manual record keeping and provides an organized way to manage farming activities and information.